Yijie Ma

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Skills

C/C++, Go, Python, OpenTelemetry (OTLP, Semantic Conventions, W3C Trace Context), Distributed Systems, gRPC, AI & Agentic Observability (GenAI, MCP, A2A), Kubernetes, API Gateway/L7 Proxy, Networking, Storage & Filesystems, Desktop App Development (Linux/macOS/Windows)

Work Experience

Google April 2017 - Present

Software Engineer

Google Cloud API Gateway & Service Infrastructure

- Led technical direction and architecture of distributed tracing for Gemini Agent across HTTP and gRPC requests, defining the OTel SemConv data model for Cloud, GenAI, Model Context Protocol (MCP), and Agent-to-Agent (A2A) requests.
- Drove cross-org alignment across Auth and Observability teams to resolve ambiguous agentic requirements; scoped milestones and mentored junior engineers on multi-quarter observability projects.
- Built end-to-end W3C Trace Context propagation, trace span generation, trace context initiation and sampling, and asynchronous OTLP trace/log export pipelines for API Gateway handling millions of QPS.
- Scoped and led multi-quarter delivery of an isolated fast API serving release track across control-plane and serving teams, decoupling rapid AI and Cloud feature rollouts from the slower stable release track.
- Pioneered reusable AI Skills to automate the release process, adopted by other team members.

gRPC Open Source Remote Procedure Call Framework

- Core developer and Maintainer of gRPC (40K+ GitHub stars) with 150+ authored PRs, driving open-source RFCs and PR reviews across EventEngine, Observability, xDS and C++ Core.
- Co-authored gRFC A79 to standardize non-per-call telemetry; architected the C++ templatized Metrics framework and OpenTelemetry plugin, cutting Counter overhead by 10%.
- Designed and built the cross-platform EventEngine asynchronous DNS resolver across Linux/POSIX and Windows.
- Designed the POSIX fork syscall handling system and resolved pre-fork deadlocks across gRPC C++ and Python.

Google Kubernetes Engine (GKE) Networking

- Engineered GKE control-plane automation in Go to auto-provision multi-NIC RDMA network topologies for AI-optimized GPU clusters.
- Designed concurrent request handling with distributed transactions and validated multi-node GPU communication using NVIDIA's NCCL test suite.

Google Drive Desktop Application

- Designed and built the rsync-style differential upload feature to save upload bandwidth for millions of users.
- Architected the USB device backup subsystem backing up millions of files daily, increasing backup speed 10x by optimizing I/O read buffers.
- Designed the cross-platform Preferences system for user and admin-level app configurations.
- Discovered a new category of crashes (1000+ daily) in threads running on heap-allocated stacks on Windows.

Stadia Cloud Gaming Service

- Designed a Linux Cgroup library for setting up container resource configurations on cloud gaming VMs.
- Increased game click-to-play launch speed by 75% (1.58s to 396ms) via CPU profiling and container setup optimization.

NetApp August 2015 - March 2017

Member of Technical Staff

Worked on the AltaVault Cloud-Integrated Storage device. Designed and built a filesystem software to backup user's files to multi-cloud vendors. Designed and implemented full and incremental snapshot backup and restore functions using RocksDB.

Education

Carnegie Mellon University Sep 2013 - May 2015

Master of Science in Electrical and Computer Engineering

Coursework: Introduction to Computer System, Embedded Realtime System, Advanced Storage System, Internet Service, Cloud Computing, Distributed System

Donghua University Sep 2009 - May 2013

Bachelor of Engineering in Electrical Engineering and Automation

Awards: 2012 IEEE-standard Micromouse competition national award issued by China Computer Federation (CCF)